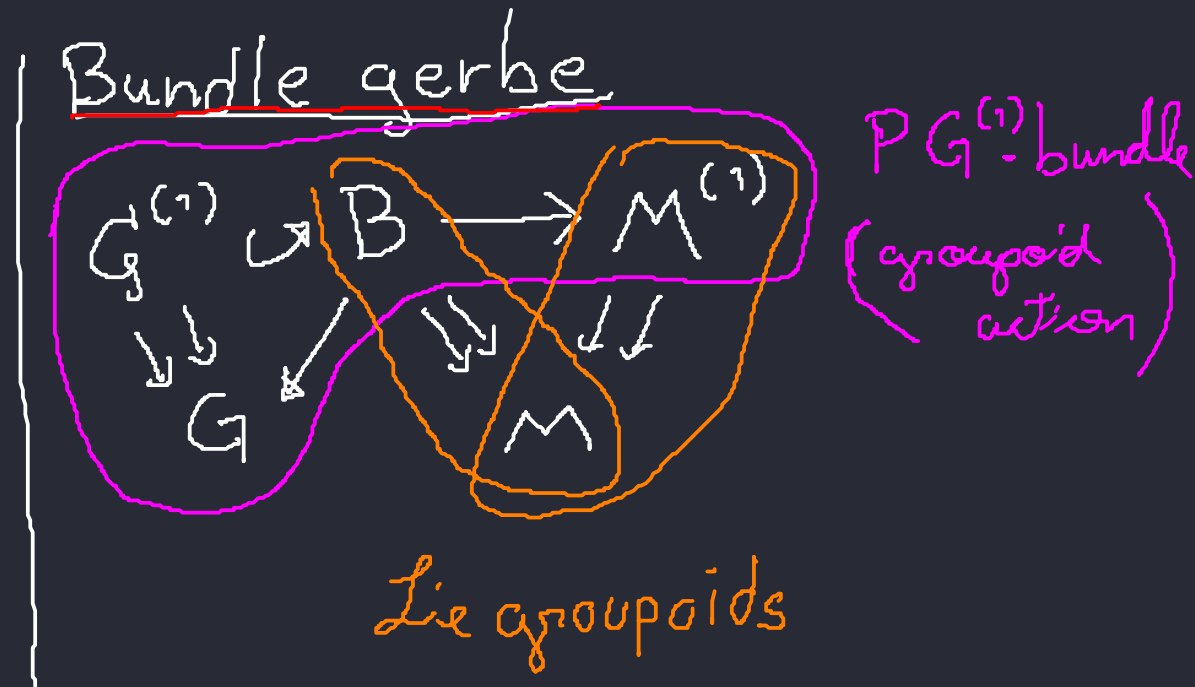
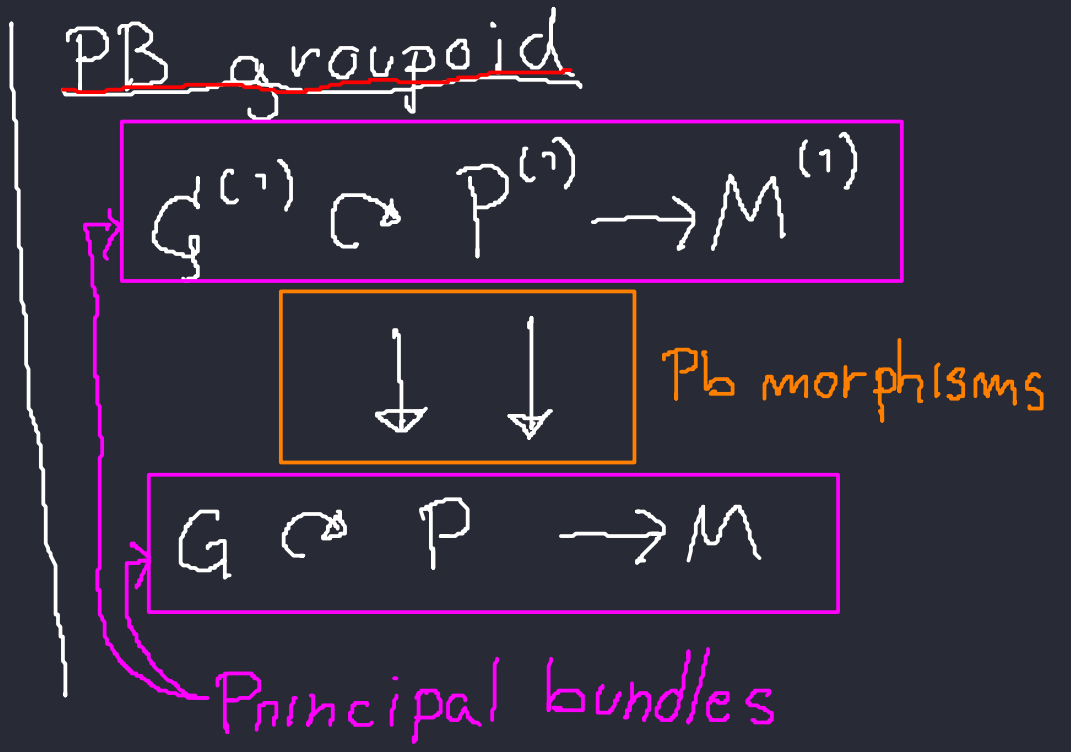
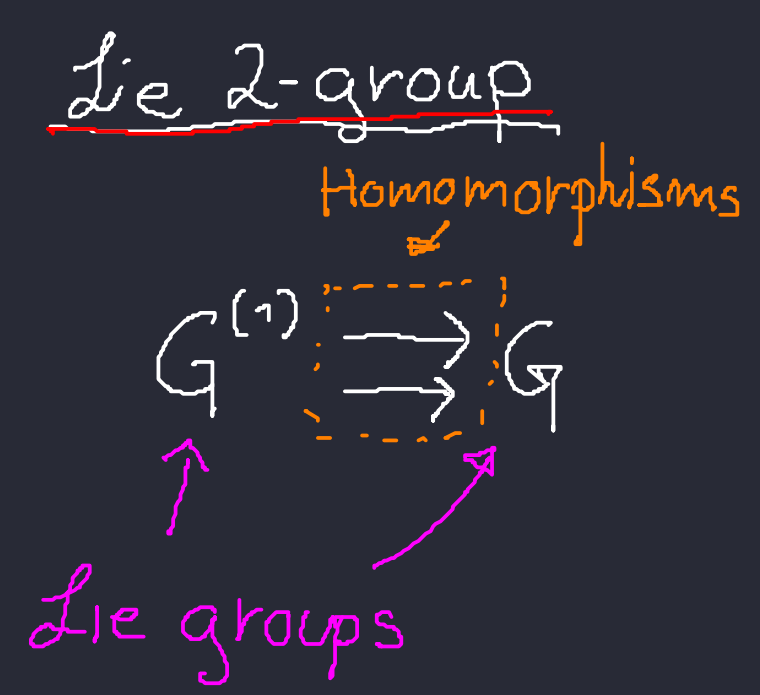


PB groupoids & bundle gerbes

Alfonso Garmendia
CRM Barcelona

Outline

- ① Lie 2-groups & examples
- ② PB groupoids
- ③ Bundle gerbes & PB groupoids



① Lie 2-Groups

$G^{(1)} \Rightarrow G \Rightarrow$
Lie groups



$\Rightarrow$ Algebra result

$$G^{(1)} \simeq \text{Ker}(S) \rtimes G$$

The map $H \rtimes G \rightarrow G^{(1)}$

$$(h, g) \mapsto h \cdot N(g)$$

is a Lie group isomorphism

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Call $H := \text{Ker}(s)$ is a Lie group

$d = t|_H: H \rightarrow G$ is homomrf

$H \curvearrowright G \quad h \cdot g = d(h) \cdot g \Rightarrow H \times G \rightrightarrows G$ Lie grd

Action grpd

① Lie 2-Groups

$G^{(1)} \rightrightarrows G \Rightarrow$
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Homomorphism

$\Rightarrow$ Algebra result

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The map $H \times G \rightarrow G^{(1)}$
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is a Lie 2group isomorphism
(Lie group & Lie groupoid iso)

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$H \curvearrowright G \quad h \cdot g = d(h) \cdot g \Rightarrow H \times G \rightrightarrows G$ Lie grp
Action grp

Prop Any Lie 2-grp is equivalent
to a "Crossed Module". i.e.

Crossed mod $\left\{ \begin{array}{l} \text{a) } d: H \rightarrow G \text{ a homomrf} \\ \text{b) } C: G \rightarrow \text{Aut}(H) \text{ (usually omitted)} \end{array} \right.$
s.t. $\forall g \in G \quad h \in H \quad h' \in H$
 $d C_g h = \alpha(dh) g^{-1} \quad \& \quad C_{dh} h' = h h' h^{-1}$

Proof ($\Rightarrow$)

$G^{(1)} \rightrightarrows G$ a Lie 2grp

$\Rightarrow H = \text{Ker}(s)$

$d = t|_H: H \rightarrow G$

$G \curvearrowright \text{Ker}(s)$ is a
Crossed Module

($\Leftarrow$)

For a Crossed Mod (G, H, d, C)

$(H \times G, \cdot) \rightrightarrows (G, \cdot)$

The action grp of
 H in G is a Lie 2-grp

Examples

① Pair groupoid
 $G \times G \rightrightarrows G$

$$k.e_1(s) = G \times \{e\} \simeq G$$

$$N: G \rightarrow G \times G, \quad g \mapsto (g, g)$$

$$* \quad G \rtimes G \simeq G \times G$$

$$(h, g) \mapsto (h, e) \cdot (g, g) = (hg, g)$$

• $G \rtimes G$ is action groupoid of G on G

$G \times G$ is the pair groupoid

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- $G \rtimes G$ is action groupoid of G on G
- $G \times G$ is the pair groupoid

② identity $G \rightrightarrows G$

③ H normal subgroup of G $H \rtimes G \rightrightarrows G$

④ given a morph $G \rightarrow \text{Gl}(V)$

the canonical maps $V \xrightarrow{d} G, v \mapsto e_G$
induce a crossed module so

$V \rtimes G \rightrightarrows G$ is a Lie 2-group

⑤ $TG \simeq \mathfrak{g} \rtimes G$ is a Lie 2-group

⑥ $\text{Pin}(V) \xrightarrow{p} O(V), O(V) \rightarrow \text{Aut}(\text{Pin}(V))$
is a crossed module

$\text{Pin}(V) \rtimes O(V) \rightrightarrows O(V)$ is a Lie 2-group

② PB groupoids

$$G^{(1)} \curvearrowright P^{(1)} \rightarrow M^{(1)}$$

↗ Lie 2-group action

Def A Lie 2-group action is a Lie grp act

$$(H \rtimes G) \times P^{(1)} \xrightarrow{a} P^{(1)} \quad \text{s.t.}$$

"a" is a Lie grpd morphism i.e.

This implies that

- $G = \{e\} \rtimes G$ acts by Lie grpd morph

- the map $\varphi: H \times P \rightarrow P^{(1)}, (h, p) \mapsto (h, e) \cdot N(p)$ satisfies

$$\underbrace{(h, g) \circ}_{\text{red}} = \underbrace{\varphi(h, g \text{ to } e)}_{\text{green}} \circ \underbrace{(g \circ)}_{\text{purple}}$$

② PB groupoids

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- the map $\varphi: H \times P \rightarrow P^{(1)}, (h, p) \mapsto (h, e) \cdot N(p)$ satisfies

$$(h, g) \circ = \varphi(h, g \text{tr}(\circ)) \circ (g \circ)$$

$$(h, g) \circ$$



$$g \circ$$



Prop if $(H \rtimes G) \hookrightarrow P^{(1)}$ is free and proper

⇒ $P^{(1)} / (H \rtimes G)$ is a Lie grpd over P/G

Examples

Given $G \curvearrowright P \xrightarrow{\pi} M$ a principal bundle

a) The identity grpd generate a PB grpd

b) Pair grpd $G^2 \curvearrowright P^{(2)} \rightarrow M^{(1)}$ is a PB grpd

c) Given $M^{(1)} \rightrightarrows M$ any Lie grpd

denote $P^{(1)} = \pi^{-1} M^{(1)} = P_{\pi} \times_t M^{(1)}_s \times_{\pi} P$

$\Rightarrow (G \times G) \curvearrowright P^{(1)} \rightarrow M^{(1)}$ is a PB grpd

and its derivative $TG \curvearrowright A_P \rightarrow A_M$ is a PB grpd

This last one is related to A_n -con and
Atiyah sequences

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d) given any surjective vector bundle map

$$W \xrightarrow{\phi} E \text{ over a mfd } M$$

$\Rightarrow V = \ker(\phi)$ is a v.b.

take $P = \text{frames}(V) \xrightarrow{\pi} M$

$\Rightarrow$ short exact sequence

$$\mathbb{R}^k \times P \simeq \pi^* V \rightarrow \pi^* W \rightarrow \pi^* E$$

$$\begin{array}{ccccc} \Rightarrow \mathbb{R}^k \rtimes G(k) & \hookrightarrow & \pi^* W & \longrightarrow & E \\ \downarrow \downarrow & & \downarrow s=t & & \downarrow s=t \\ G(k) & \hookrightarrow & P & \longrightarrow & M \end{array}$$

③ Bundle gerbes & PB grps

A bundle gerbe is

$$\begin{array}{ccc} H \rtimes G \hookrightarrow B^{(1)} & \rightarrow & M^{(1)} \\ \downarrow \downarrow & \swarrow \searrow & \downarrow \downarrow \\ G & & M \end{array}$$

• Principal
(groupoid-action)
bundle

• grpd morph

s.t. $\forall k_i \in H \rtimes G \quad b_i \in B$

$$\Rightarrow (k_1 * b_1) \circ (k_2 * b_2) = (k_1 \cdot k_2) * (b_1, b_2)$$

$$\begin{array}{ccc} H \rtimes G \hookrightarrow G \times B & \rightarrow & M^{(1)} \\ \downarrow \downarrow & \downarrow \downarrow & \downarrow \downarrow \\ G \hookrightarrow G \times M & \rightarrow & M \end{array} \quad \begin{array}{l} \text{a PB} \\ \text{grpd} \end{array}$$

$$s(g, b) = (g, s(b)) \quad t(g, b) = (s^P(b), t(b))$$

$$(h, g) \cdot (g', b) = (g g', (c_{g(b)} g'^{-1} g^{-1} h, b) * b)$$

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$$s(g, b) = (g, s(b)) \quad t(g, b) = (sP(b), t(b))$$

$$(h, g) * (g', b) = (gg', (c_{P(b)} g'^{-1} g^{-1} h, b) * b)$$

$$\begin{array}{ccc} H \rtimes G & \hookrightarrow & P^{(1)} \rightarrow M^{(1)} \\ \downarrow & \downarrow & \downarrow \\ G & \hookrightarrow & G \times M \rightarrow M \end{array} \quad \text{Inverse functor}$$

$$\Rightarrow P^{(1)} \simeq G \times \underset{B}{\text{Ker}(P_{G \circ S})} \quad \text{differs}$$

$\Rightarrow B$ acquires a groupoid str $P^{(1)}/G$

$$\Rightarrow P^{(1)} \simeq G \times B$$

It works for morphisms

$$\text{Mor}(P^{(1)}) = (H \rtimes G) \text{ equivariant Maps } P^{(1)} \rightarrow H \rtimes G$$

$$\downarrow \\ \text{Mor}(B) = H \rtimes G \text{ equivariant Maps } P^{(1)} \rightarrow H$$